

PRACTICAL

WEEK 2

2520090111

SATWIKI GANDHE

```
1 • CREATE DATABASE bookflow_db;
2 • USE bookflow_db;
3 • CREATE TABLE Books(
4     book_id int primary key,
5     title varchar(100) not null,
6     isbn varchar(20) unique,
7     published_year int check(published_year<2027)
8 );
9 • INSERT INTO Books(book_id, title, isbn, published_year) Values
10 (1,'the great gatsby','9780743273565', 1925),
11 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
12 (3, '1984', '9780451524935', 1949);
13 • SELECT * FROM Books;
14
15 • CREATE TABLE Members(
16     member_id int primary key,
17     full_name varchar(100),
18     email varchar(100) unique
19 );
```

```

7   published_year int check(published_year<2027)
8   );
9   • INSERT INTO Books(book_id, title, isbn, published_year) Values
10  (1,'the great gatsby','9780743273565', 1925),
11  (2, 'To Kill a Mockingbird', '9780061120084', 1960),
12  (3, '1984', '9780451524935', 1949);
13  • SELECT * FROM Books;

```

Result Grid | Filter Rows: | Edit: | Export/Import

	book_id	title	isbn	published_year
▶	1	The Alchemist	9780061122415	1988
	2	Clean Code	9780132350884	2008
	3	Atomic Habits	9780735211292	2018
*	NULL	NULL	NULL	NULL

Result Grid
Form Editor
Field Types

```

18  email varchar(100) unique
19  );
20  • insert into Members(member_id, full_name, email) values
21  (101, 'john smmith', 'john.smith@email.com'),
22  (102, 'emma wilson', 'emmma.wilson@email.com'),
23  (103, 'michael brown', 'micheal.brown@email.com');
24  • SELECT * FROM Members;

```

Result Grid | Filter Rows: | Edit: | Export/Import

	member_id	full_name	email
▶	1	Anil Kumar	anil.kumar@example.com
	2	Priya Sharma	priya.sharma@example.com
	3	Ravi Verma	ravi.verma@example.com
	101	john smmith	john.smith@email.com
	102	emma wilson	emmma.wilson@email.com
	103	michael brown	micheal.brown@email.com
*	NULL	NULL	NULL

Result Grid
Form Editor
Field Types

Members 2 x Apply Revert

```

25
26 • create table Loans(
27     loan_id int primary key,
28     member_id int,
29     book_id int,
30     loan_date date,
31     foreign key(member_id) references members(member_id),
32     foreign key(book_id) references books(book_id)
33 );
34
35 • INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
36     (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
37     (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
38     (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
39     (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
40     (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
41 • SELECT * FROM Loans;
42
43

```

Limit to 1000 rows

```

35 • INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
36     (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
37     (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
38     (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
39     (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
40     (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
41 • SELECT * FROM Loans;

```

Result Grid

	loan_id	member_id	book_id	loan_date
▶	1	101	1	2025-01-05
	2	102	2	2025-01-08
	3	103	3	2025-01-10
	4	101	2	2025-02-01
	5	102	1	2025-02-05
	6	103	2	2025-02-12
	7	101	3	2025-03-01
	8	102	3	2025-03-07
	9	103	1	2025-03-15
	10	101	1	2025-04-01

Loans 3 x

Apply Revert Co

Output

Limit to 1000 rows

```
44 • SELECT m.full_name AS Member_Name, b.title AS Book_Title
45 FROM Loans l
46 INNER JOIN Members m ON l.member_id = m.member_id
47 INNER JOIN Books b ON l.book_id = b.book_id;
48
49 • SELECT published_year, COUNT(book_id) AS Total_Books
50 FROM Books
```

Result Grid

	Member_Name	Book_Title
▶	john smmith	The Alchemist
	emma wilson	The Alchemist
	michael brown	The Alchemist
	john smmith	The Alchemist
	emma wilson	Clean Code
	john smmith	Clean Code
	michael brown	Clean Code
	michael brown	Atomic Habits
	john smmith	Atomic Habits
	emma wilson	Atomic Habits

Result Grid
Form Editor
Field Types

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48
49 • SELECT published_year, COUNT(book_id) AS Total_Books
50 FROM Books
51 GROUP BY published_year
52 ORDER BY published_year;
53
```

Result Grid

	published_year	Total_Books
▶	1988	1
	2008	1
	2018	1

Result Grid
Form Editor
Field Types

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53
54 • CREATE TABLE Donation_History (
55     donation_id INT PRIMARY KEY,
56     book_id INT,
57     donor_name VARCHAR(100),
58     donation_date DATE,
59     FOREIGN KEY (book_id) REFERENCES Books(book_id)
60 );
61
62 • START TRANSACTION;
63
64 • INSERT INTO Books (book_id, title, isbn, published_year)
65 VALUES (4, 'Animal Farm', '9780451526342', 1945);
66
67 • INSERT INTO Donation_History (donation_id, book_id, donor_name,
68 VALUES (1, 4, 'Raj Kumar', CURDATE());
69
70 • COMMIT;
```

[Conti](#)

Output

Action Output

	#	Time	Action	Message
✓	14	21:50:34	SELECT published_year, COUNT(book_id) AS To...	3 row(s) returned
✓	15	21:50:50	CREATE TABLE Donation_History (donation_...	0 row(s) affected
✓	16	21:50:53	START TRANSACTION	0 row(s) affected
✓	17	21:50:58	INSERT INTO Books (book_id, title, isbn, publish...	1 row(s) affected
✓	18	21:51:01	INSERT INTO Donation_History (donation_id, bo...	1 row(s) affected
✓	19	21:51:05	COMMIT	0 row(s) affected

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68 VALUES (1, 4, 'Raj Kumar', CURDATE());
69
70 • COMMIT;
71
72
73 • CREATE INDEX idx_books_isbn ON Books(isbn);
74 • SELECT * FROM Books WHERE isbn = '9780451524935';
```

Result Grid Filter Rows: Edit: Export/Import

	book_id	title	isbn	published_year
•	NULL	NULL	NULL	NULL

Result Grid
Form Editor
Field Types

Books 7 Apply Revert Conte

Output				
Action Output				
	#	Time	Action	Message
✓	16	21:50:53	START TRANSACTION	0 row(s) affected
✓	17	21:50:58	INSERT INTO Books (book_id, title, isbn, publish...	1 row(s) affected
✓	18	21:51:01	INSERT INTO Donation_History (donation_id, bo...	1 row(s) affected
✓	19	21:51:05	COMMIT	0 row(s) affected
✓	20	21:51:22	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates
✓	21	21:51:25	SELECT * FROM Books WHERE isbn = '978045...	0 row(s) returned